Data (Section 3.1)

GWFSS v1.0, labelled set:
1,096 RGB images from
10 institutions
Classes: background, head,
stem, leaf
Stem: 3.8% of pixels;
modal, semantic labels

Splits (Section 3.2)

Random split: 70 / 10 / 20%
per institution (in-distribution)

Region split: 7 institutions
for training, UTokyo for
validation, UQ for testing

Leave one institution out
(LOIO): 10 folds, each
institution tested once

Training (Sections 3.3–3.5)

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{Dice}} + \lambda\,\mathcal{L}_{\mathrm{conn}} \quad \text{Eq. (1)}$$

$\mathcal{L}_{\mathrm{conn}}$ acts on the stem probability only:

- **none** — baseline
- **clDice** — soft skeletons, Eqs. (2)–(3)
- **Skeleton Recall** — reference skeleton tube, Eq. (4)
- **Tversky control** — recall-weighted, no skeleton, Eq. (5)

Architectures: DeepLabV3+ (ResNet-101), SegFormer-B1 and -B2, UPerNet (ConvNeXt-T); 8,000 iterations; final iterate; $\lambda = 1$ unless stated

Experiment	Runs	Seeds
Random split, 4 arch. × 3 losses	36	3
Region split, 4 arch. × 3 losses	36	3
×2 resolution, 2 arch. (+ clDice k = 40)	18 + 6	3
LOIO, 2 arch. × 3 losses	60	1
LOIO baselines, second seed	20	1
LOIO Tversky control, 2 arch.	20	1
LOIO λ = 0.25, 0.5, SegFormer-B2, 4 folds	16	1
Total	**212**	

Inference and calibration (3.6, 3.8)

Class decision: arg max over the four logits
Threshold calibration: offset b added to the stem logit before the arg max, Eq. (10);
b chosen on the validation institution or on k = 5 or 10 labelled target images (20 random draws); no retraining

Evaluation (Sections 3.7–3.9)

Pixels: IoU per class; stem precision, recall and area ratio A, Eqs. (6)–(7)

Connectivity: clDice and skeleton recall scores, Betti-0 error; merged, split, spurious and missed components, Eq. (12)

Traits: visible stem length, stem area, fragments; relative error, Eq. (8)

Paired per-image tests; bootstrap intervals over images or institutions, Eqs. (9), (11)